



PYXIS

Pyxis: Leading the sustainable maritime revolution with scalability and real-time data

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GOOGLE CLOUD RESULTS

- ✓ Significantly reduces infrastructure costs
- ✓ Enables rapid scalability and speeds up time to market
- ✓ Improves real-time monitoring, enhancing vessel safety and operational efficiency
- ✓ Increases collaboration across distributed teams, boosting productivity and innovation

By moving to Google Cloud and leveraging the flexibility and scalability of Google Kubernetes Engine, Pyxis reduced operational costs, and increased its agility and security, which supported

the expansion of its fleet and tech solutions.

“Initially, we focused on hiring software infrastructure developers with maritime expertise. However, our collaboration with Niveus and Google Cloud demonstrated that cross-domain experience can introduce valuable and innovative ideas.”

Jin Yi Low
Chief Innovation Officer,
Pyxis

The maritime industry is exploring opportunities for advancements in efficiency and sustainability. The sector often relies on diesel-powered vessels, which contribute to environmental pollution and incur high operational costs. There is a growing need for solutions that can optimize energy consumption and reduce emissions. In this environment, companies must adopt advanced technologies to stay competitive and meet stringent regulatory requirements.

[Pyxis](#) is a forward-thinking tech company in the maritime industry dedicated to creating sustainable and innovative solutions for shipping and vessel management. With its electric vessels and energy-efficient fleet management system, Electra™, Pyxis aims to lead the industry toward a greener future.

However, the company’s existing infrastructure was insufficient to meet the demands of real-time data transmission and monitoring for their vessels. "We needed a fast and reliable way to publish certain data attributes like system or equipment status or alerts to the cloud for real-time monitoring," says Jin Yi Low, Chief Innovation Officer at Pyxis.



Maintaining local servers would have required substantial resources and a larger in-house team, and diverted valuable time and workforce away from innovation and core business activities. Jin Yi Low says, "We initially considered managing everything on local premises on a small scale, but we quickly recognized that to achieve rapid growth without significant funding, we needed to adopt an asset-light approach and leverage cloud services."

Initially, Pyxis sought a software infrastructure solution with specific maritime capabilities. Yet, they quickly recognized that having robust technological expertise and innovation capabilities were more important. "Initially, we focused on hiring software infrastructure developers with maritime expertise. However, our collaboration with Niveus and Google Cloud demonstrated that cross-domain experience can introduce valuable and innovative ideas," says Low.

Pyxis chose Google Cloud for its advanced capabilities and alignment with the company's green mission to reduce carbon emissions, optimize energy consumption, and promote more environmentally-friendly practices.

Pyxis leverages the flexibility of [Google Kubernetes Engine \(GKE\)](#) to manage and deploy containerized applications across its various environments, ensuring seamless software updates and scalability. Meanwhile, [Pub/Sub](#) facilitates real-time data transmission from its vessels to the cloud, enabling onshore personnel to monitor and maintain the vessels proactively.

Support from Niveus was also crucial in facilitating a smooth transition. They helped Pyxis significantly reduce infrastructure costs, allowing Pyxis to focus on innovation and growth.



Driving efficiency and security in operations

One of Pyxis' flagship products, Pyxis One, is an electric launch vessel that operates within harbor ecosystems, providing crew transfers and shipments between land and ocean-going vessels. Pyxis uses GKE to deploy its backend services and frontend applications. This setup ensures that updates and new features can be rolled out with no interruptions, maintaining high levels of efficiency and reliability.

Additionally, Pyxis uses Pub/Sub for real-time data transmission. This real-time monitoring capability enables proactive maintenance and immediate response to potential issues, ensuring the vessels operate at optimal efficiency.

“Thanks to Google Cloud, our team can now handle the cloud infrastructure easily. It's amazing how the right cloud solution can boost our operational efficiency and allow us to focus on other avenues for innovation.”

Jin Yi Low
Chief Innovation Officer,
Pyxis

To ensure the security of its operations, Pyxis uses [Cloud Armor](#) to scan for malicious incoming requests and safelist traffic based on specific parameters and domains. This adds a robust layer of security, protecting sensitive data and maintaining the integrity of the systems. [Cloud SQL](#) provides a managed database solution that stores all operational data securely. With Cloud SQL, Pyxis can scale its data storage needs effortlessly, accommodating the growing volume of data generated by its expanding fleet of electric vessels.

Pyxis leverages [Cloud Build](#) and [Artifact Registry](#) to compile, test, and create service images. [Cloud Logging](#) and [Cloud Monitoring](#) under the Operations Suite are used to debug and monitor its infrastructure and services. These tools provide detailed insights into system performance, helping the team to identify and resolve issues quickly.

With Google Cloud, Pyxis has significantly reduced infrastructure costs and streamlined its operations. This has allowed the company to focus more on innovation and less on managing local servers and hardware. Streamlining their process has cut Pyxis' time to market in half, allowing them to rapidly deploy new features and updates to their fleet.

"Thanks to Google Cloud, our team can now handle the cloud infrastructure easily. It's amazing how the right cloud solution can boost our operational efficiency and allow us to focus on other avenues for innovation."

The transition to Google Cloud has also facilitated a more agile development environment. The support from [Niveus](#) has been instrumental in this process. Their expertise has helped Pyxis respond quickly to changing requirements and market conditions.

"At Niveus, we take pride in delivering cloud solutions that create a meaningful impact for our clients. Partnering with Pyxis has allowed us to contribute to their visionary mission of steering the maritime industry toward a more sustainable future with the power of Google Cloud. Witnessing the transformative results of our collaboration has been truly gratifying, and we are deeply committed to supporting Pyxis' continued growth and success," adds Bharath Yaragatti, Vice President - Global Expansion, Niveus Global.



Sailing toward a smarter, greener future with AI

"We're keeping a close watch on AI. Google Cloud

As part of its ongoing commitment to innovation, Pyxis is exploring the use of artificial intelligence (AI) to improve operational efficiency

AI could significantly improve our service offerings and open up new revenue models.”

Jin Yi Low

Chief Innovation Officer,
Pyxis

and drive sustainable practices.

One of the key areas where Pyxis plans to implement AI is predictive maintenance. By analyzing historical data from the vessels, AI models can predict potential failures before they occur, allowing for timely interventions and reducing downtime. "We need AI to accurately pinpoint demand and availability, especially as we transition from diesel to electric vessels," says Low.

By leveraging Google Cloud's advanced AI and machine learning tools, Pyxis aims to gain deeper insights into their operations and make data-driven decisions. This will further improve their operational efficiency and contribute to the company's sustainability goals.

Moving forward, Pyxis plans to expand its AI capabilities to include more sophisticated data analysis and automation. The company believes that AI-driven insights will become integral to their strategic planning and day-to-day operations. Low states, "We're keeping a close watch on AI. Google Cloud AI could significantly improve our service offerings and open up new revenue models.”

Pyxis is a forward-thinking maritime tech company dedicated to creating sustainable and innovative solutions for shipping and vessel management. They develop advanced technologies that optimize energy consumption and efficiency and manufacture electric vessels, like their flagship craft, the Pyxis One, with zero emissions and low maintenance requirements.

Industry: Maritime

Location: Singapore

Products: [Google Cloud](#), [Google Kubernetes Engine](#), [Cloud Pub/Sub](#), [Artifact Registry](#), [Cloud Armor](#), [Cloud Build](#), [Cloud Monitoring](#), [Cloud SQL](#), [Cloud Trace](#), [Compute Engine](#)

About Google Cloud Partner: Niveus

Niveus is a leading cloud consulting firm and a trusted Google Cloud partner dedicated to helping businesses accelerate their digital transformation. Focusing on innovation and excellence, Niveus develops customized cloud solutions that drive operational efficiency and sustainable growth. With a collaborative approach at heart, they deliver tailored solutions that meet the unique challenges and goals of each client.

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